

Q2 Oral Exam

Weight of assessment in final grade: 20%

Type of assessment: individual

Evaluator: Supervisors

The students are evaluated based on their ability to discuss how their project and process relates to the criteria below.

Domain	Learning Outcome	Criteria	4-5 Insufficient	6 Sufficient	7-8 Good	9-10 Excellent	Comments
Communicate	1.c. Demonstrate understanding of the scientific publishing process.	Scientific publishing process	Limited or incorrect understanding. Confuses steps or shows little awareness of how scientific publishing works.	Basic overview of the process. Understands submission and review in general terms but lacks clarity or depth.	Good understanding of key steps. Aware of peer review and journal selection but may lack detail on some aspects.	Clear and accurate understanding of the full publishing process, including cover letter, journal selection, and revisions/peer review.	
Collaborate	2.a. Develop partnerships with other groups, stakeholders and scientists.	Knowledge on teamwork: - Roles of partnerships - Benefits of partnerships	The student does understand the benefits of partnership or have any idea how to build them	The student has basic understanding of the collaboration-related processes.	The student has detailed knowledge on roles and benefits of collaboration	The student has shown they can build partnerships and collaborations	
	2.b. Process management: plan, schedule and execute the research project.	Process management	The student is unable to describe how the group developed the plan.	The student is able to describe basic group process of developing their plan.	The student is able to explain the choices they made clearly about their planning	The student excels in describing how the plan meets needs and desires of group members.	
	2.c. Identify problems and formulate solutions to collaboration-related problems	Problems and solutions in collaboration	The student does not recognize key collaboration issues or propose meaningful solutions.	The student identifies basic problems and offers simple or partially effective solutions.	The student clearly identifies relevant problems and proposes well-reasoned, feasible solutions.	The student insightfully identifies complex collaboration issues and formulates effective, proactive solutions.	
	2.d. Understand ethics and standard practice around credit and ownership in scientific group work.	Credit and ownership	The student does not know about credit, ownership and group work	The student has basic understanding of credit, ownership and group work	The student as detailed knowledge of credit, ownership and group work	The student has detailed knowledge of credit, ownership and group work and can apply these in reflecting on their own experiences within the team and their own behaviour.	
Research	3.a Develop research	Research strategy	The student	The student partially	The student effectively	The student excels in	

	plan and gather the right sources and knowledge to complete their project.		demonstrates a limited ability to describe the development of research project plan. The strategy is unclear, lacks depth, or does not adequately consider diverse sources and perspectives.	describes an effective strategy for developing a research project but it may lack clarity or depth in considering multiple sources, perspectives, and methodologies.	develops a clear and coherent strategy for a research project and methodologies. The strategy demonstrates an understanding of the importance of diverse sources and perspectives in the research process.	developing an exceptionally well-detailed and comprehensive strategy for doing a research project The strategy demonstrates a deep understanding of how to approach their project using multiple sources, perspectives, and methodologies.	
	3.b. Critically evaluate research strategies, and if necessary, modify and adapt these strategies.	Research strategy reflection	The student demonstrates a limited ability to critically evaluate their plan and adapt it as needed	The student partially thinks critically about their methodology but may resist changes or not see large problems.	The student effectively updates their strategy and shows critical thinking and flexibility in adapting to challenges and opportunities.	The student excels in improving and adapting their project and thinking critically about strategies.	
	3.c. Plan and (if possible) conduct an experiment based on their research	Research Planning	Limited understanding of the research requirements and how to implement their experiment.	Partial understanding of experimental requirements, with some recognition of contexts, magnitude, connections, time planning and collaboration. May overlook key aspects.	Effective development of an experiment plan, demonstrating a clear grasp of the research steps necessary.	Exceptional development of realistic experimental plan even if unable to complete the research.	
	3.d. Analyse outcomes of an experiment in a critical way	Data analysis understanding	Limited understanding of data analysis	Partial understanding of data analysis and management	Effective development of data analysis and management plan. Made good decisions but cannot explain why	Exceptional development of a data analysis and management plan, is able to describe why they made the decisions they did.	
	3.e Apply research ethics around reproducibility, data management and integrity.	Ethics in research	Did not show understanding of ethical issues in reproducibility, data management and integrity.	Struggles with identifying specific problems in own work or others work.	Understands problems, is able to discuss understanding of ethics in reproducibility, data management and integrity.	Understands problems, is able to discuss understanding of ethics in reproducibility, data management and integrity providing interesting insight.	

	3.f Understand processes involved with institutional reviews, grant applications and scientific publication	Research review processes	Does not demonstrate understanding of scientific review processes			Understands processes of scientific review including research integrity, grants and publications	
	3.g Apply ethical and philosophical considerations in performing research.	Philosophy and ethics	Does not demonstrate understanding of ethical and philosophical considerations in research.			Identifies problems, has long-term vision, and reflects on ethical and philosophical context of research.	
Reflect	4.b. Ask for constructive feedback of peers.	Feedback	FAIL (0 pts): Student has not asked for feedback these past ten weeks or acted on the feedback given.		PASS (10 pts): Student has asked for feedback these past ten weeks or has acted on the feedback given.		
	4.c. Understand how personal biases influence collaboration.	Personal biases	FAIL (0 pts): Student shows no or poor understanding of biases in relation to collaboration and science.		PASS (10 pts): Student shows understanding of biases in relation to collaboration and science.		
	4.d. Reflect on transdisciplinary collaboration in such a way that it can improve collaboration.	Transdisciplinary collaboration	FAIL(0 pts): The student provides little or no reflection, or fails to connect it to improving collaboration. Is unwilling to take ownership of their role.		PASS (10 pts): The student reflects meaningfully on transdisciplinary collaboration and suggests insights or actions that could enhance future collaboration.		